

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2025-2026

Mid-Semester Test
EC-2 Regular

Course No. : AIMLCZC416
Course Title : Mathematical Foundations for Machine Learning
Nature of Exam : Closed Book
Weightage : 30%
Duration : 2 Hours
Date of Exam : 21-12-2025

No. of Pages = 2

No os questions = 5

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.
4. Notation used here are as per the text book.

Q1. Consider the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & -1 & 0 \end{bmatrix}$.

(a) Solve the system of equations $AX = b$, where $b = \begin{bmatrix} 6 \\ 12 \\ 5 \end{bmatrix}$ by finding the Echelon form of the augmented matrix. [3M]

(b) Find the Echelon form of $B = \begin{bmatrix} 5 & 25 & -64 & 32 \\ 0 & -1 & -3 & 4 \\ 0 & 1 & -2 & 5 \\ 0 & 0 & 5 & -3 \end{bmatrix}$. Hence find the determinant of B . Also find the rank of B . [2M]

Q2. Consider the vectors $\begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} 4 \\ -8 \\ 5 \end{bmatrix}$ in R^3 .

(a) Write $\begin{bmatrix} 4 \\ -8 \\ 5 \end{bmatrix}$ as linear combination of the vectors $\begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix}$. Explain how this will imply the vectors $\begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} 4 \\ -8 \\ 5 \end{bmatrix}$ are linearly dependent in R^3 . [3M]

(b) Prove that the $V = \text{span} \left\{ \begin{bmatrix} 5 \\ -3 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ -3 \end{bmatrix} \right\}$ is a subspace of R^3 . Find a basis and dimension of V . [2M]

(c) Is the element $\begin{bmatrix} 4 \\ -8 \\ 2 \end{bmatrix}$ in V which is defined in (b). Justify your answer. [1M]

Q3. (a) Find the Singular Value decomposition of $A = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$. [4M]

(b) Is the matrix $A = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$ diagonalizable? Justify your answer. [2M]

Q4. Consider the matrix $A = \begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix}$.

(a) Define an inner product on R^2 using the matrix A by $\langle X, Y \rangle = X^T A Y$. What are the properties of A that ensures that $\langle X, Y \rangle = X^T A Y$ is an inner product. Prove those properties. [3M]

(b) Using this inner product find the distance between the vectors $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$. [2M]

(c) Verify whether the vectors $\begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ are orthogonal with respect to this inner product. [1M]

Q5. Consider the function

$$f(x, y) = x^2 + xy + 2y^2 - 4x + 3.$$

(a) Compute the gradient $\nabla f(x, y)$. [2M]

(b) Using (a) find the critical point(s) of f . [1M]

(c) Determine whether the critical point is a point of local minimum, local maximum, or saddle point.

[2M]

(d) Let $g(x) = f(x, 1)$. Find the Taylor series expansion of $g(x)$ at $x = 1$. [2M]
